NEURAL INFORMATION PROCESSING SYSTEMS

Object-aware Contrastive Learning for Debiased Scene Representation

Sangwoo Mo[*1], **Hyunwoo Kang**[*1], Kihyuk Shon[2], Chun-Liang Li[2], Jinwoo Shin[1]

* Equal contribution

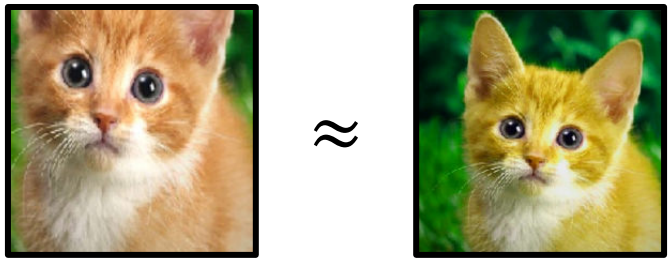
[1]KAIST [2]Google Cloud AI





Background: Self-supervised learning for images

- Self-supervised learning (SSL) have shown remarkable success on image domain[1,2]
  - **Key idea:** Enforce invariance against different augmentations of the same sample



[1] He et al. Improved Baselines with Momentum Contrastive Learning. Tech Report 2020.
[2] Grill et al. Bootstrap your own latent: A new approach to self-supervised Learning. NeurIPS 2020.

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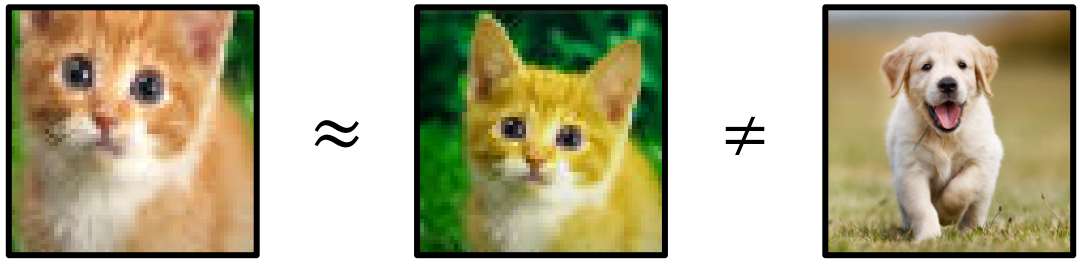
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  - Specifically,

= positives = negatives

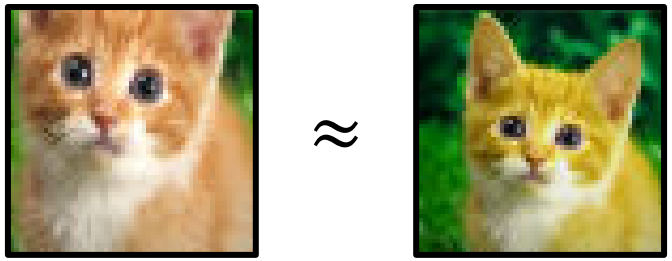
    - Contrastive learning (e.g., MoCov2[1]) pulls the same samples (of diff. augs.) while pushes diff. samples



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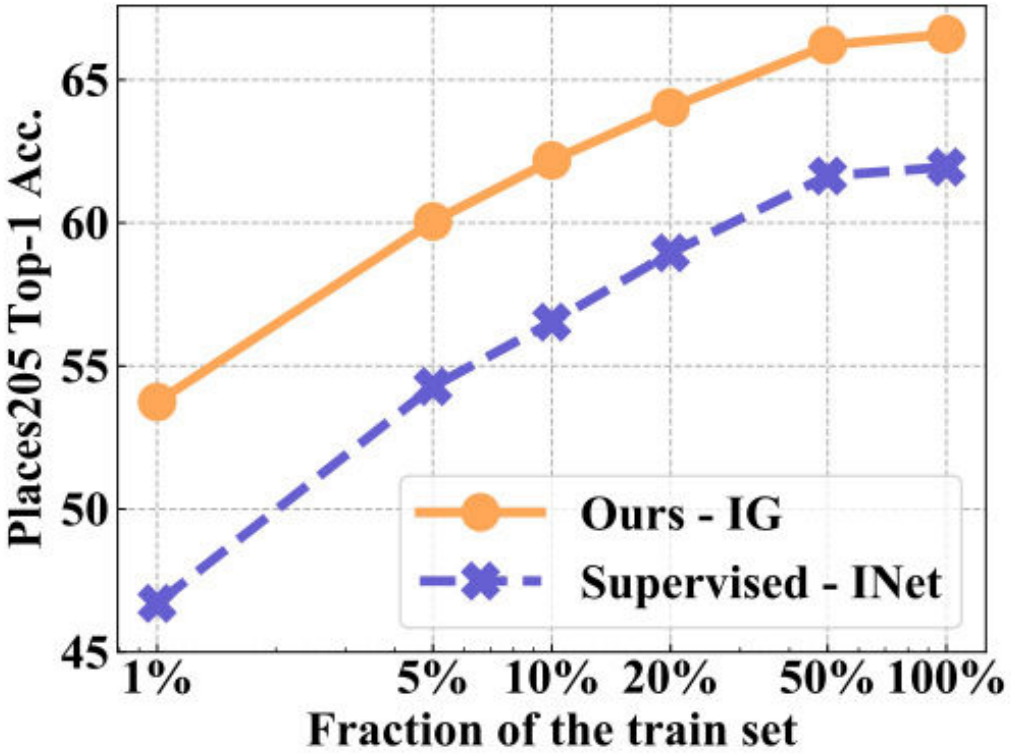
= positives = negatives

    - Positive-only methods (e.g., BYOL[2]) only pulls the same samples (i.e., no negatives)

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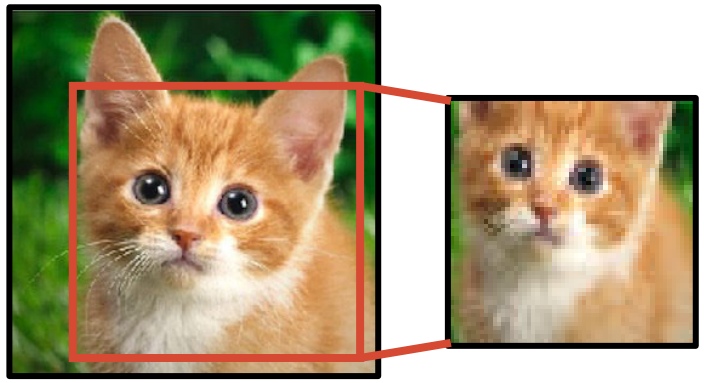
- Self-supervised learning (SSL) have shown remarkable success on image domain
  - Trained on large-scale unlabeled data, SSL shows phenomenal performance on various tasks[1]
    - Provides a better backbone than the ImageNet pre-trained model



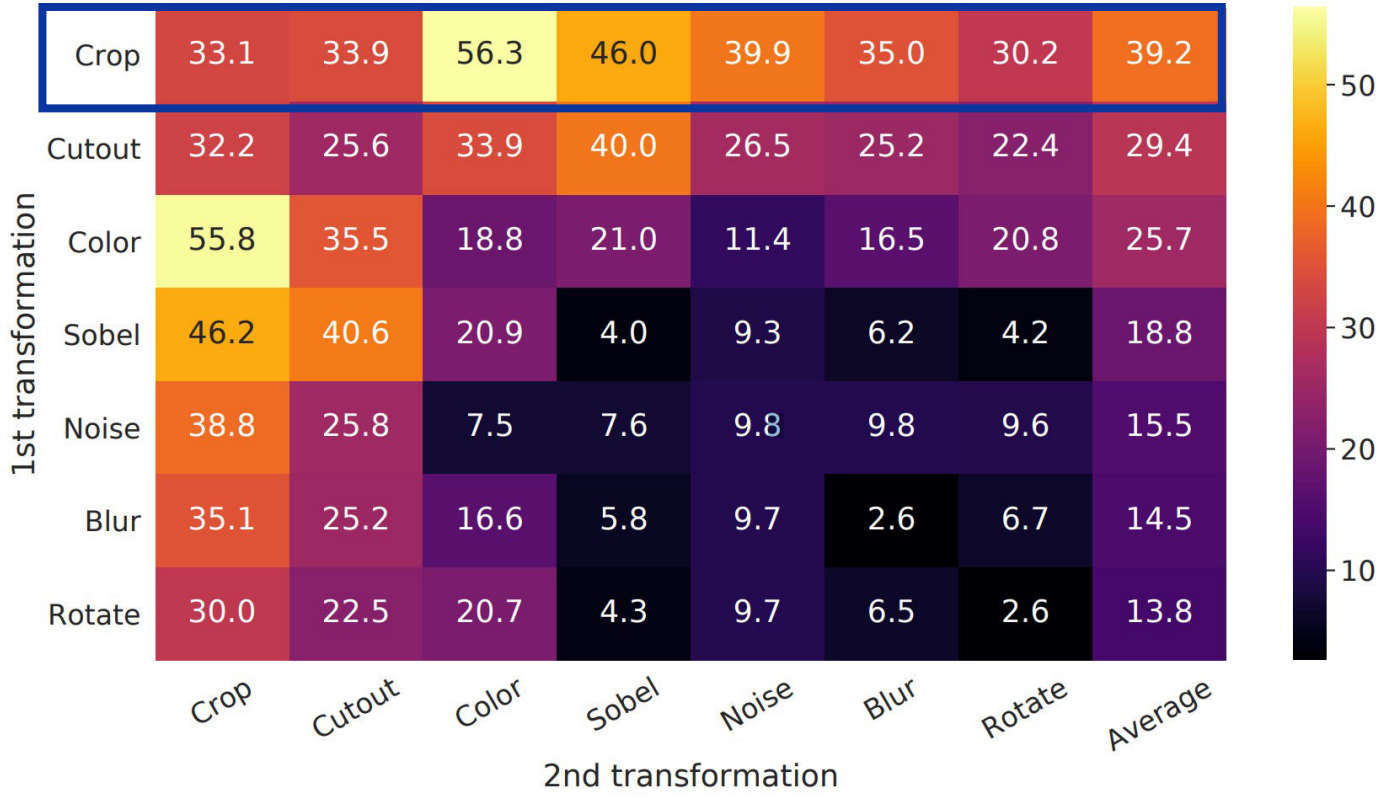
[1] Goyal et al. Self-supervised Pretraining of Visual Features in the Wild. Tech Report 2020.

Motivation: Scene bias of visual self-supervised learning

- Visual SSL heavily relies on the random crop augmentation[1]
  - Assume that the cropped views represent the same sample



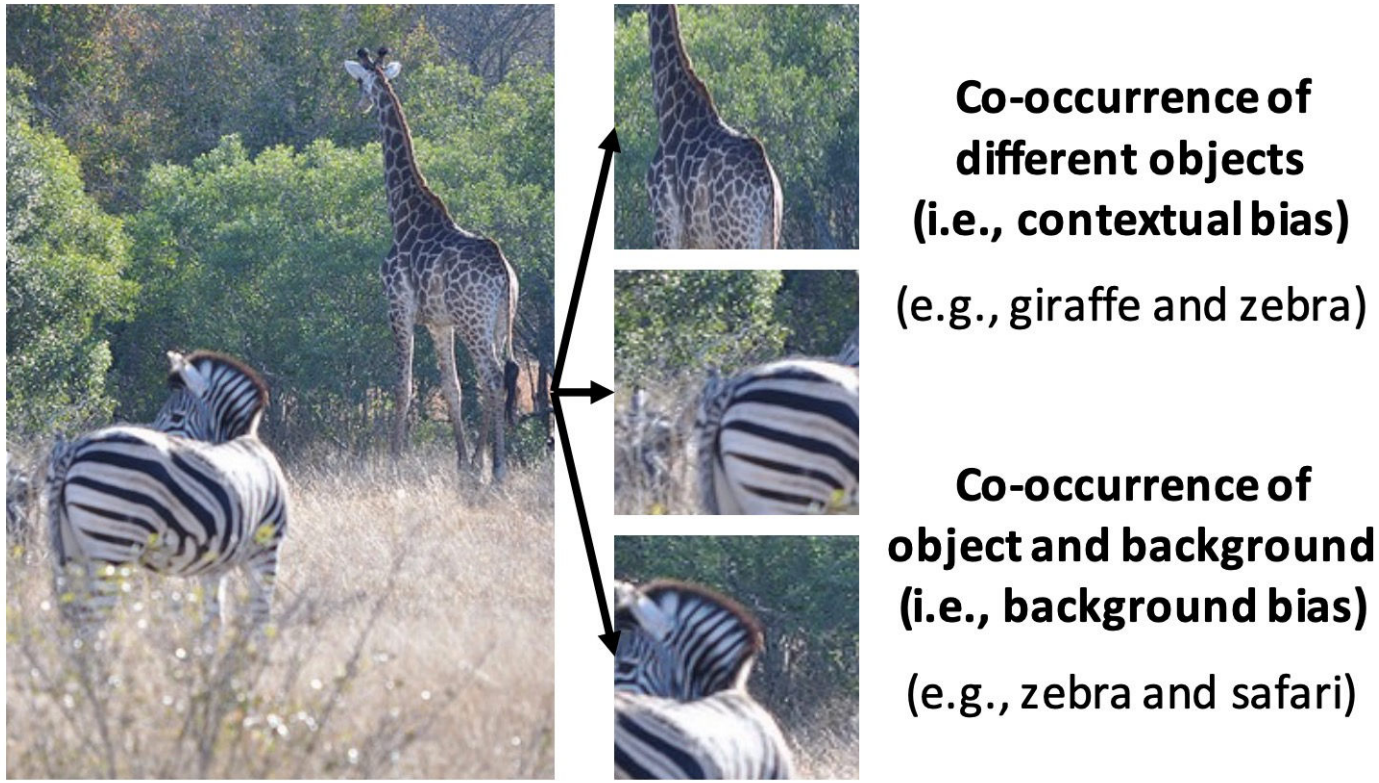
Random Crop



[1] Chen et al. A Simple Framework for Contrastive Learning of Visual Representations. ICML 2020.

Motivation: Scene bias of visual self-supervised learning

- However, entangling the randomly cropped patches occur the **scene bias**
  - **Contextual bias:** Entangle the features of different objects (for multi-object images)
  - **Background bias:** Entangle the features of object and background



Co-occurrence of different objects (i.e., contextual bias)

(e.g., giraffe and zebra)

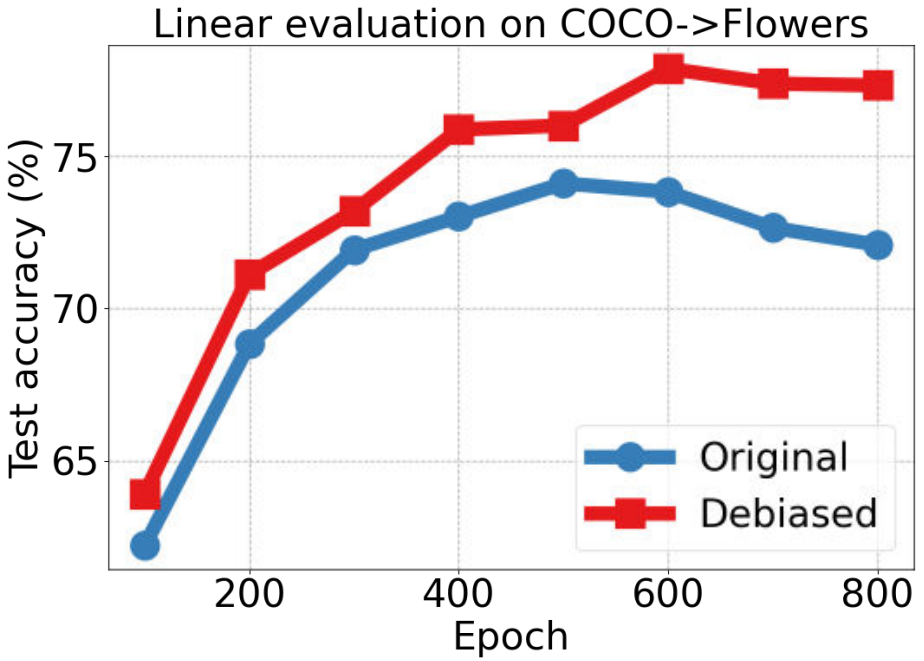
Co-occurrence of object and background (i.e., background bias)

(e.g., zebra and safari)

Motivation: Scene bias of visual self-supervised learning

- However, entangling the randomly cropped patches occur the **scene bias**

  - **Contextual bias:** Entangle the features of different objects (for multi-object images)
    - It harms the discriminative power of representation (i.e., performance drop on downstream tasks)
      - **+1:** The original model often even loses its discriminator power as training goes



MoCov2

Motivation: Scene bias of visual self-supervised learning

- However, entangling the randomly cropped patches occur the **scene bias**
  - **Background bias:** Entangle the features of object and background
    - It harms the generalization on background shifts (look at the background instead of the object)



Original class: **Bird**
Predicted class: **Fish**

Motivation: Scene bias of visual self-supervised learning

- However, entangling the randomly cropped patches occur the **scene bias**
  - **Background bias:** Entangle the features of object and background
    - It harms the generalization on background shifts (look at the background instead of the object)
      - **+1:** Debiasing it also improves the generalization on other distribution shifts (e.g., shape bias, corruption)



Original class: **Bird**
Predicted class: **Fish**



(a) ImageNet-Sketch [29] (b) Stylized-ImageNet [39] (c) ImageNet-R [40] (d) ImageNet-C [30]

Method: Object-aware data augmentations

- We propose object-aware data augmentations to reduce the scene bias
  - **Reducing contextual bias:** Object-aware random crop (avoid entangle of diff. objects)
  - **Reducing background bias:** Background mixup (disentangle object and background)
  - Both augmentations are applied before other augmentations of self-supervised learning

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  - However, both augmentations need object information

    ⇒ How to infer the object information in an **unsupervised** manner?

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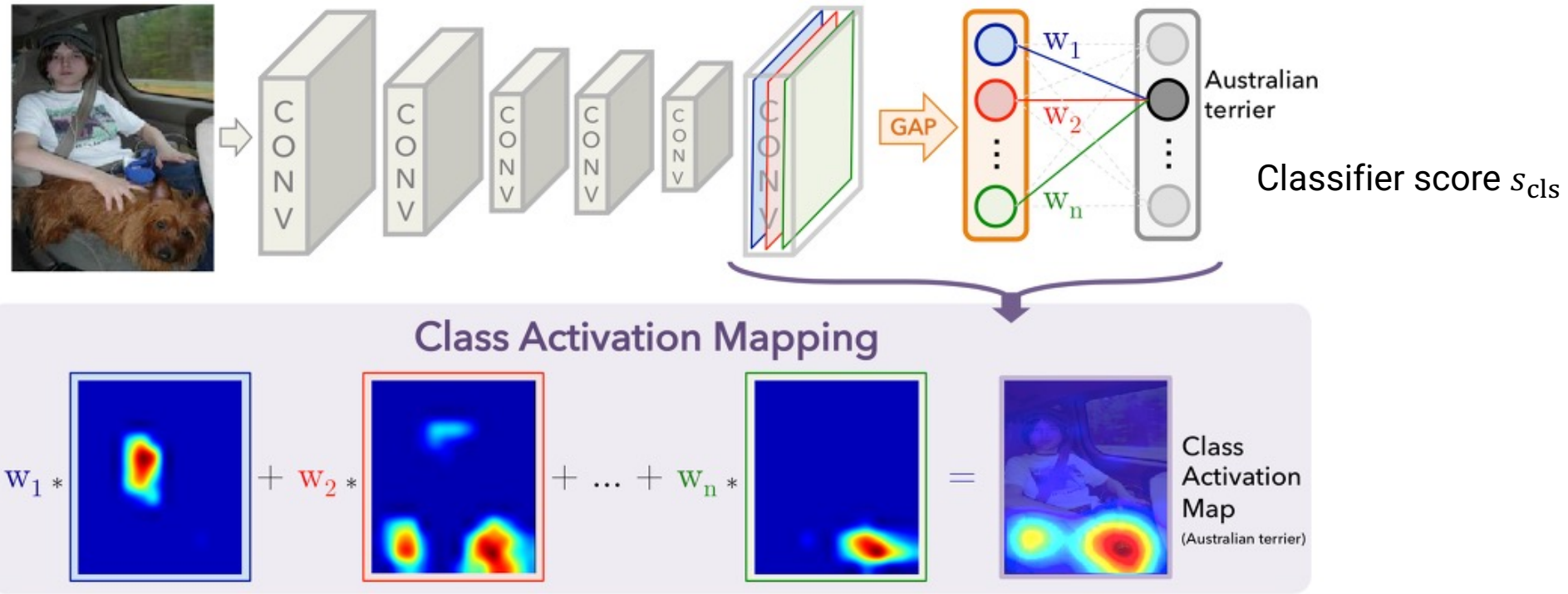
- However, both augmentations need object information

  ⇒ How to infer the object information in an **unsupervised** manner?

To this end, we also propose a simple yet effective object localization method which can be inferred from a contrastively learned representation for free

Method: Contrastive Class Activation Map (ContraCAM)

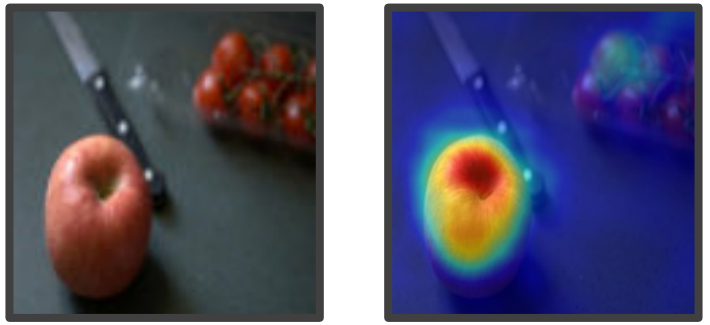
- **Background:** Gradient-based Class Activation Map (Grad-CAM)[1]
  - Find the region that is most salient for classification
  - Define the saliency map as a weighted sum of the penultimate activations



[1] Selvaraju et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV 2017.

Method: Contrastive Class Activation Map (ContraCAM)

- Instead, ContraCAM finds the most discriminative region of x compared to other images $\mathcal{B}$
  - I.e., change the classifier score $s_{\text{cls}}(x)$ to the contrastive score $s_{\text{con}}(x; \mathcal{B})$
  - Note that the most discriminative region of an image is often the object



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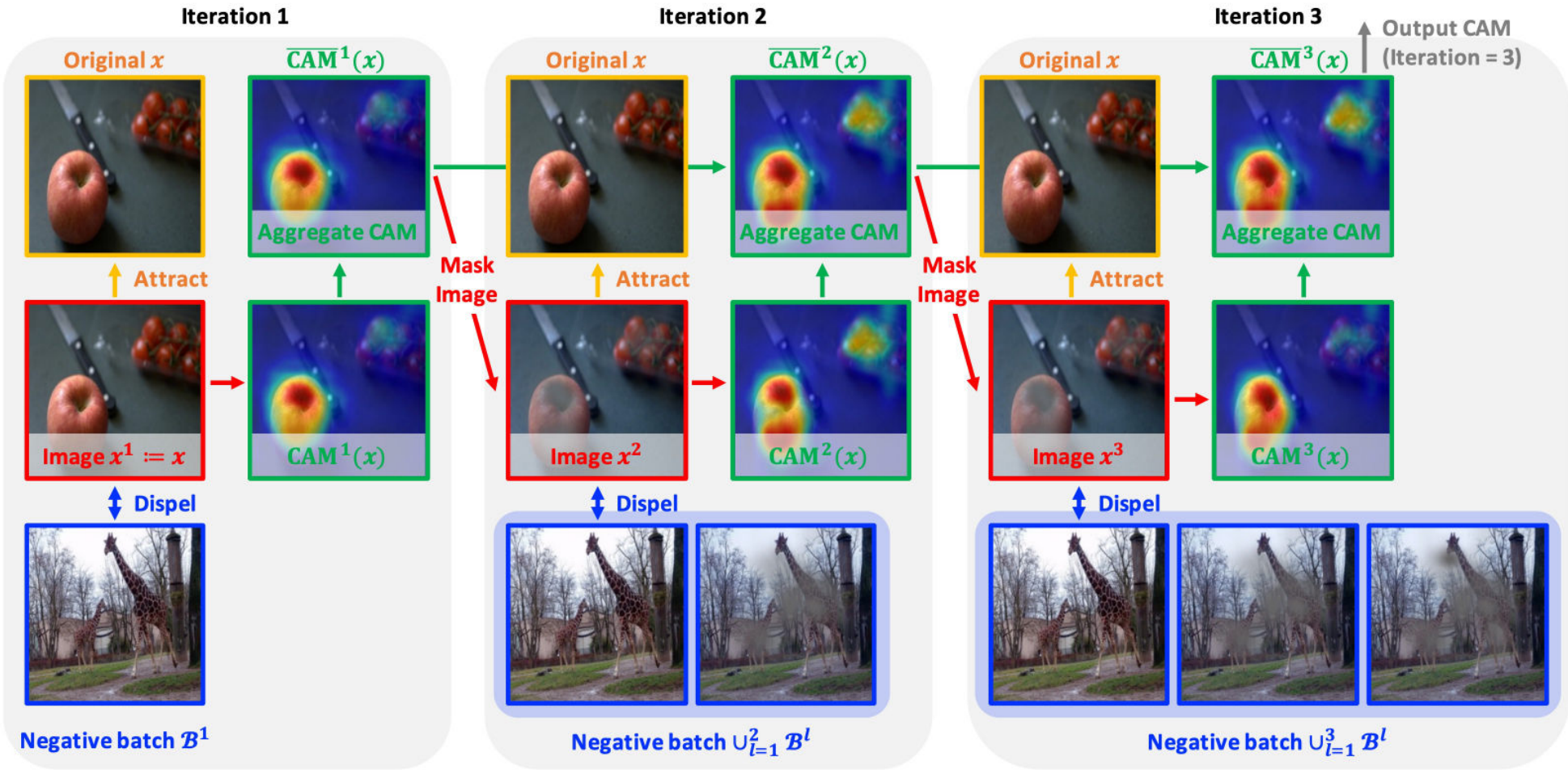
- Instead, ContraCAM finds the most discriminative region of x compared to other images $\mathcal{B}$
  - I.e., change the classifier score $s_{\text{cls}}(x)$ to the contrastive score $s_{\text{con}}(x; \mathcal{B})$
  - Here, we found that a trick of discarding negative signals is crucial to the performance

$$\texttt{CAM}_{ij} = \texttt{Normalize}\left(\texttt{ReLU}\left(\sum_k \alpha_k A_{ij}^k\right)\right), \quad \alpha_k = \texttt{ReLU}\left(\frac{1}{HW}\sum_{i,j}\frac{\partial s_{\text{con}}(x; x, \mathcal{B} \setminus x)}{\partial A_{i,j}^k}\right)$$

Red color highlights the differences from the classifier CAM

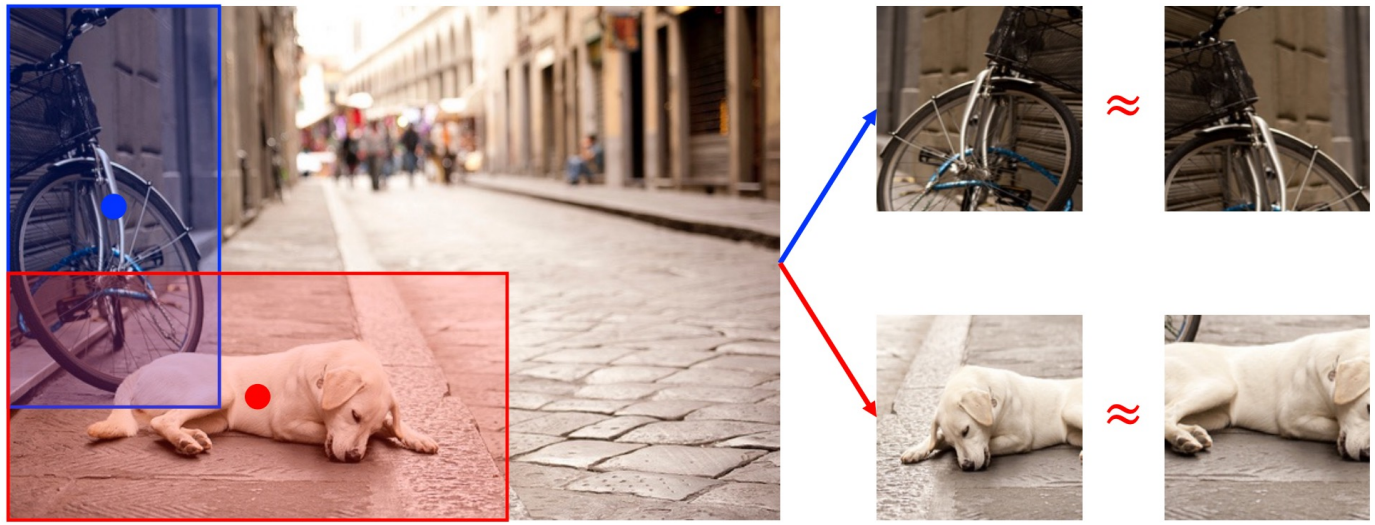
Method: Contrastive Class Activation Map (ContraCAM)

- Instead, ContraCAM finds the most discriminative region of x compared to other images $\mathcal{B}$
  - We further improve the localization results via an iterative procedure



Method: Object-aware Random Crop (OA-Crop)

- OA-Crop enforces the random crop performed inside the object box
  - The object box is inferred by ContraCAM
  - It avoids the entanglement of different objects (unless boxes are intersected)

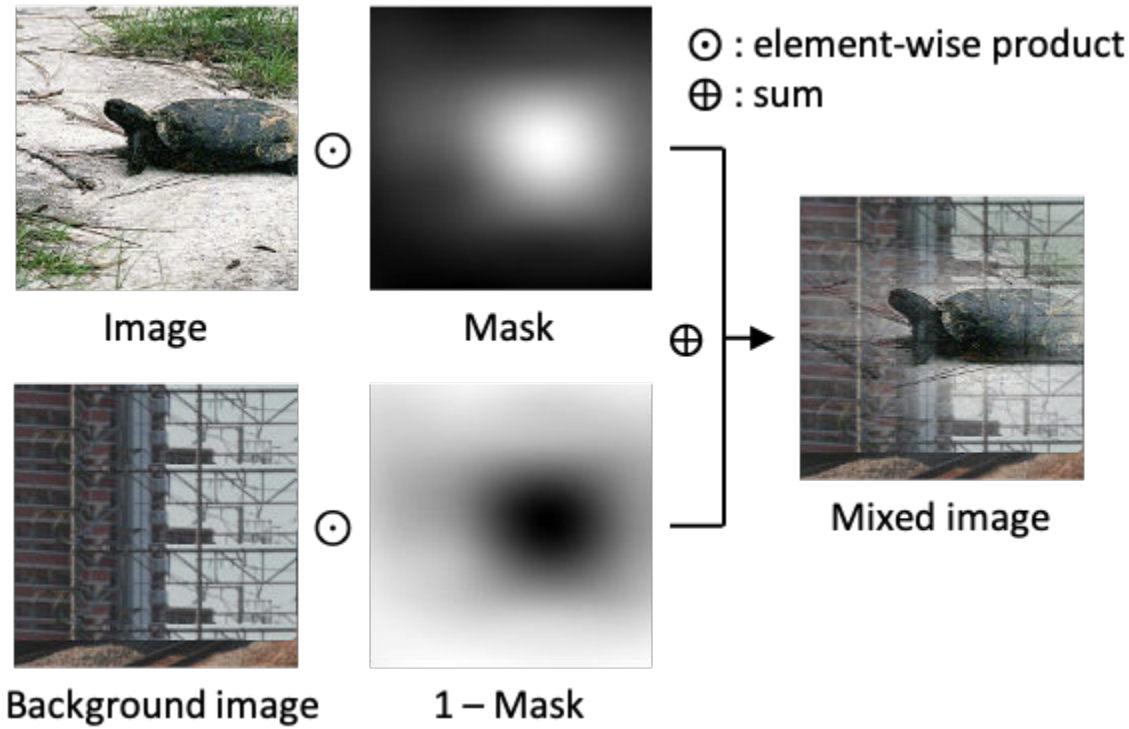


```
if len(boxes) > 0: # can be empty
    box = random.choice(boxes)
    image = image.crop(box)
# apply other augmentations (e.g., random crop)
```

Only a few line additions of code

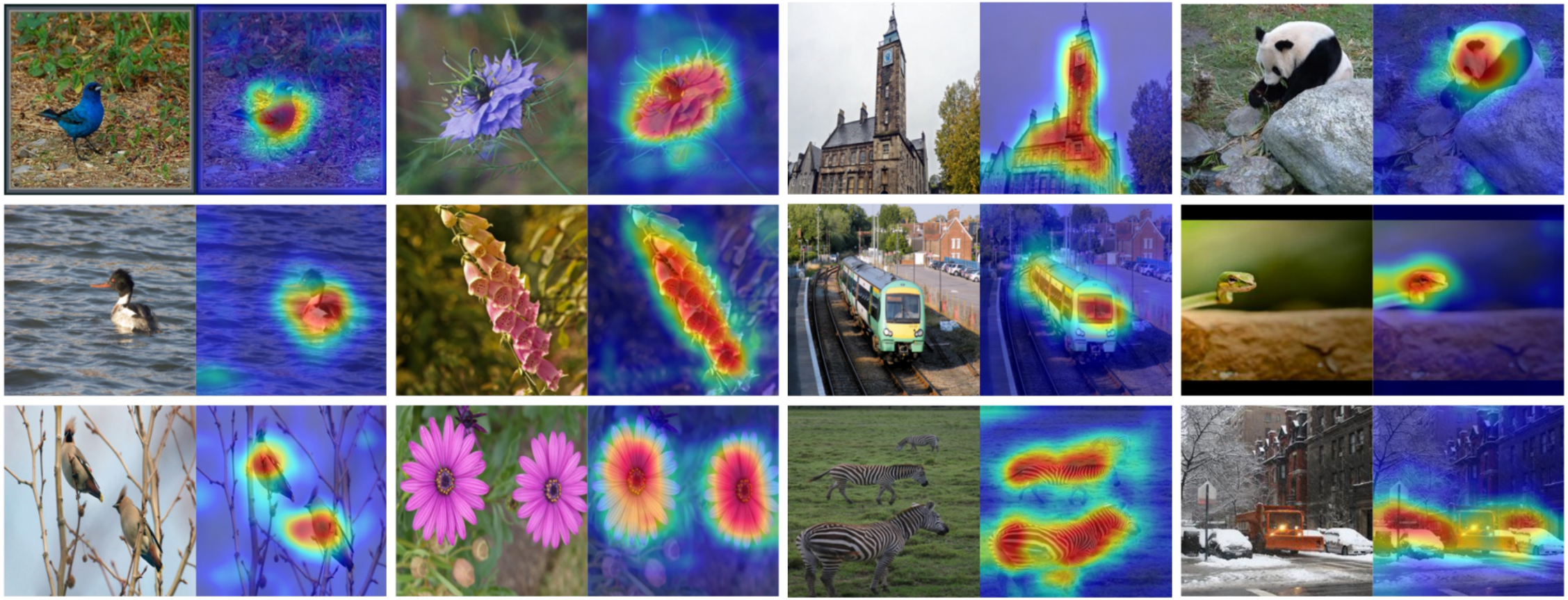
Method: Background Mixup (BG-Mixup)

- BG-Mixup mixes the object of image #1 and background of image #2
  - The mask of image #1 is inferred by ContraCAM
  - Intuitively, the mixed image disentangles the background from the object



Results: Unsupervised object localization (via ContraCAM)

- ContraCAM detects objects well on various datasets



(a) CUB [46] (b) Flowers [26] (c) COCO [25] (d) ImageNet-9 [23]

* We train a MoCov2 for each dataset to infer ContraCAM

Results: Unsupervised object localization (via ContraCAM)

- ContraCAM is comparable to the SOTA unsupervised object localization method (ReDO[1])

Method	CUB	Flowers	COCO	ImageNet-9
ReDO [50]	0.426	0.764	0.286	0.416
ContraCAM w/o negative signal removal	0.287	0.555	0.242	0.361
ContraCAM (ours)	**0.460**	**0.776**	**0.319**	**0.427**

- …and also comparable to the classifier CAM using the ImageNet-trained classifier

Model	Method	ImageNet	CUB	Flowers	VOC	OpenImages
Supervised	Classifier CAM	55.95	55.52	76.87	53.88	48.01
MoCov2	ContraCAM (ours)	55.88	64.07	75.64	59.40	49.89

[1] Chen et al. Unsupervised Object Segmentation by Redrawing. NeurIPS 2019.

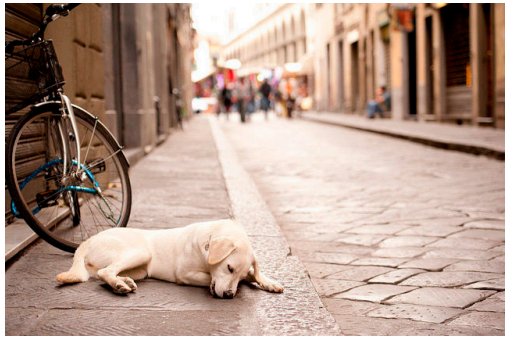
Results: Setup for object-aware self-supervised learning

- For OA-Crop and BG-Mixup experiments,
  1. We first train vanilla MoCov2/BYOL and infer ContraCAM using them
  2. We then train MoCov2/BYOL applied with (ContraCAM-based) OA-Crop/BG-Mixup from scratch
     - We also compare with OA-Crop/BG-Mixup using ground-truth (GT) object information

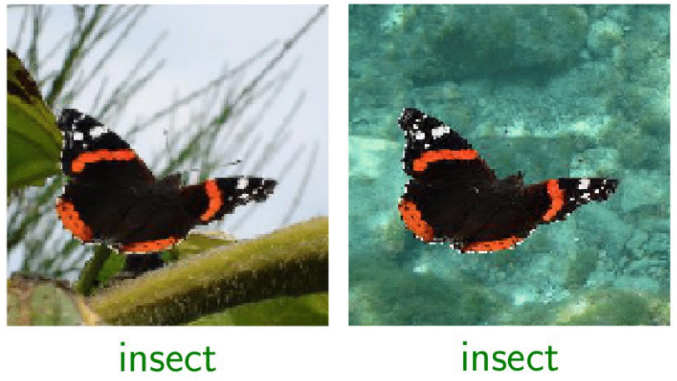
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- We use
  - COCO dataset for OA-Crop experiments



  - Background Challenge dataset for BG-Mixup experiments



Results: Learning from multi-object images (via OA-Crop)

- OA-Crop improves the discriminative power of the learned representations
  - Linear evaluation on various downstream tasks

Model	Network	OA-Crop	Test dataset						
			COCO-Crop	CIFAR10	CIFAR100	CUB	Flowers	Food	Pets
MoCov2	ResNet-50	-	74.30	77.58	53.26	22.90	72.09	59.70	59.25
MoCov2	ResNet-50	CAM	76.37 (+2.07)	84.10 (+6.52)	62.72 (+9.46)	25.46 (+2.56)	77.33 (+5.24)	62.01 (+2.31)	60.97 (+1.72)
MoCov2	ResNet-50	GT	76.44 (+2.14)	84.03 (+6.45)	62.81 (+9.55)	22.59 (-0.31)	75.09 (+3.00)	57.47 (-2.23)	57.67 (-1.58)
BYOL	ResNet-50	-	73.36	76.62	51.79	21.95	73.77	59.49	60.72
BYOL	ResNet-50	CAM	74.92 (+1.56)	82.79 (+6.17)	61.13 (+9.34)	24.34 (+2.39)	77.83 (+4.06)	61.83 (+2.34)	61.27 (+0.55)
BYOL	ResNet-50	GT	80.69 (+7.33)	85.92 (+9.30)	65.06 (+13.27)	28.68 (+6.73)	77.95 (+4.18)	64.63 (+5.14)	65.69 (+4.97)
MoCov2	ResNet-18	-	67.38	66.83	41.85	15.36	58.81	45.88	45.37
MoCov2	ResNet-18	CAM	69.92 (+2.54)	76.73 (+9.90)	53.25 (+11.40)	16.26 (+0.90)	64.77 (+5.96)	48.56 (+2.68)	47.37 (+2.00)
MoCov2	ResNet-18	GT	71.60 (+4.22)	77.99 (+11.16)	53.32 (+11.47)	18.19 (+2.83)	65.43 (+6.62)	46.41 (+0.53)	48.68 (+3.31)
BYOL	ResNet-18	-	67.74	67.82	41.96	17.24	64.79	49.58	52.90
BYOL	ResNet-18	CAM	70.85 (+3.11)	77.37 (+9.55)	54.79 (+12.83)	18.24 (+1.00)	70.56 (+5.77)	53.16 (+3.58)	54.27 (+1.37)
BYOL	ResNet-18	GT	76.59 (+8.85)	81.23 (+13.41)	58.11 (+16.15)	22.99 (+5.75)	73.25 (+8.46)	55.33 (+5.75)	59.80 (+6.90)

* Interestingly, MoCov2 + OA-Crop (GT) often degrade the performance. This is because the GT objects often contain **not discriminative** objects, which makes **contrasting them difficult**. It does not happen for the OA-Crop (ContraCAM) since ContraCAM only selects the discriminative regions.

Results: Learning from multi-object images (via OA-Crop)

- OA-Crop also improves the generalization on distribution shifts
  - On par for natural images (ImageNet-9) but more robust for shape-biased and corrupted samples

Model	Crop	Test dataset				
		ImageNet-9	ImageNet-Sketch-9	Stylized-ImageNet-9	ImageNet-R-9	ImageNet-C-9
MoCov2	Baseline	84.67	41.44	18.94	32.40	26.08
MoCov2	OA-Crop (CAM)	84.54 (–0.13)	43.11 (+1.68)	20.50 (+1.56)	32.35 (–0.05)	27.85 (+1.77)
MoCov2	OA-Crop (GT)	82.49 (–2.18)	46.85 (+5.42)	22.18 (+3.24)	33.68 (+1.28)	26.81 (+0.73)
BYOL	Baseline	84.07	44.28	17.91	32.13	27.51
BYOL	OA-Crop (CAM)	84.67 (+0.60)	45.05 (+0.77)	20.21 (+2.29)	32.64 (+0.51)	28.70 (+1.19)
BYOL	OA-Crop (GT)	86.72 (+2.65)	51.52 (+7.25)	22.95 (+5.03)	36.28 (+4.15)	31.65 (+4.14)

* Used ResNet-50 for these results

Results: Generalization on distribution shifts (via BG-Mixup)

- BG-Mixup improves the generalization on background shifts
  - Blue/red arrows imply higher/lower is better (i.e., more rely on the objects than background)

	MoCov2			BYOL		
Dataset	Baseline	BG-Mixup (CAM)	BG-HardMix (GT)	Baseline	BG-Mixup (CAM)	BG-HardMix (GT)
Original ↑	89.17±0.49	90.73±0.05 (+1.56)	89.69±0.14 (+0.52)	87.30±0.61	89.30±0.02 (+2.00)	90.95±0.33 (+3.65)
Only-BG-B ↓	31.29±2.46	29.60±0.89 (-1.69)	26.44±1.63 (-4.85)	25.59±0.78	25.70±3.46 (+0.11)	27.28±0.04 (+1.69)
Only-BG-T ↓	44.91±0.16	41.95±0.38 (-2.96)	40.11±0.58 (-4.80)	42.83±0.51	39.94±0.52 (-2.89)	41.16±0.17 (-1.67)
Only-FG ↑	63.62±4.71	70.55±1.71 (+6.93)	72.68±0.69 (+9.06)	61.04±0.94	67.53±0.30 (+6.49)	72.63±1.13 (+11.59)
Mixed-Same ↑	80.98±0.34	84.13±0.33 (+3.15)	84.48±0.17 (+3.50)	79.30±0.31	81.28±0.53 (+1.98)	84.94±0.47 (+5.64)
Mixed-Rand ↑	60.34±0.66	66.89±0.54 (+6.55)	71.95±0.54 (+11.61)	58.03±0.85	63.83±0.53 (+5.80)	70.51±0.33 (+12.48)
Mixed-Next ↑	55.50±0.71	63.64±0.41 (+8.14)	70.25±0.14 (+14.75)	53.35±0.36	63.05±3.54 (+9.70)	66.81±0.08 (+13.46)
BG-Gap ↓	20.64±0.36	17.24±0.31 (-3.40)	12.53±0.69 (-8.11)	21.27±0.64	17.45±0.15 (-3.82)	14.44±0.56 (-6.83)

* Used ResNet-18 for these results

Results: Generalization on distribution shifts (via BG-Mixup)

- BG-Mixup also improves the generalization on other distribution shifts
  - Better than Mixup[1] and CutMix[2] for shape-biased and corrupted samples

Model	Augmentation	Test dataset			
		ImageNet-Sketch-9	Stylized-ImageNet-9	ImageNet-R-9	ImageNet-C-9
MoCov2	Baseline	46.70±0.67	25.66±0.54	37.51±0.80	31.82±0.40
MoCov2	+Mixup [41]	51.18±0.88 (+4.48)	32.36±0.12 (+6.70)	41.00±0.12 (+3.49)	40.15±2.07 (+8.33)
MoCov2	+CutMix [42]	45.92±0.88 (-0.78)	26.46±0.68 (+0.80)	37.07±0.31 (-0.44)	32.29±0.60 (+0.47)
MoCov2	+BG-Mixup (ours)	**52.15**±0.93 (+5.45)	**33.36**±0.61 (+7.70)	**41.50**±0.45 (+3.99)	**44.39**±0.89 (+12.57)
BYOL	Baseline	45.15±1.12	23.80±0.45	36.21±0.31	28.62±0.06
BYOL	+Mixup [41]	50.12±1.61 (+1.97)	**28.11**±1.15 (+4.31)	37.90±0.44 (+1.69)	32.48±0.55 (+3.86)
BYOL	+CutMix [42]	46.07±0.05 (+1.39)	23.98±0.05 (+0.18)	35.43±0.44 (-0.78)	29.68±0.39 (+1.06)
BYOL	+BG-Mixup (ours)	**52.40**±0.70 (+7.25)	27.01±0.74 (+3.21)	**39.62**±0.21 (+3.41)	**33.83**±0.28 (+5.21)

* Used ResNet-18 for these results

[1] Zhang et al. mixup: Beyond Empirical Risk Minimization. ICLR 2018.
[2] Yun et al. CutMix: Regularization Strategy to Train Strong Classifiers with Localizable Features. ICCV 2019.

Summary

- We propose an **object-aware self-supervised** learning framework:
  - **ContraCAM:** Unsupervised object localization using contrastively learned representations
  - **OA-Crop:** Debias contextual bias → Learning from multi-object images
  - **BG-Mixup:** Debias background bias → Generalization on distribution shifts

- See more details:
  - **Paper:** https://arxiv.org/abs/2108.00049
  - **Code:** https://github.com/alinlab/object-aware-contrastive

Thank you for your attention! 😇